



A COMMON INTRODUCTION

McCode-vibing

Experiences using standard LLM models for writing instruments etc.

McStas — neutrons

McXtrace — X-rays

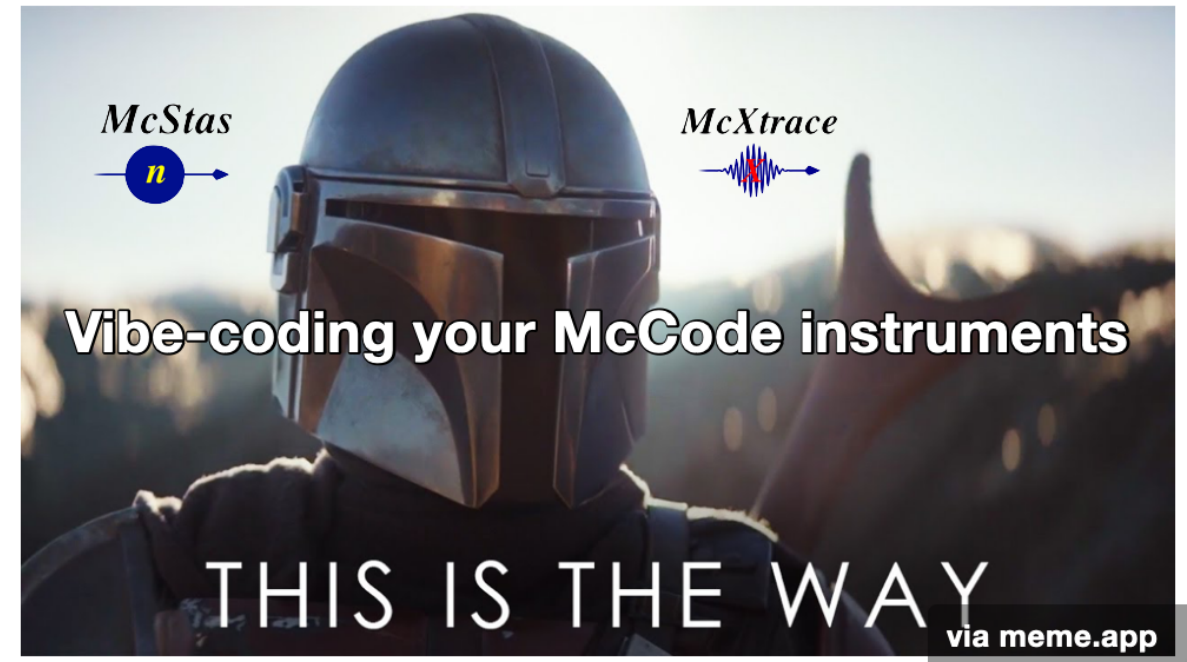


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Examples of vibe-coding with McCode

- 1 “Build me a powder diffractometer ”
- 2 “Use information on the web, draft me an instr”
- 3 Full use-case from Anders Kästner, PSI





“Build me a powder diffractometer ”

Prompt:

Based on available information in <https://mcstas.org> and <https://github.com/mccode-dev/McCode/wiki>, please put together the code for a diffractometer at a reactor source.

Here are the main characteristics

1. Input parameter lambda (wavelength), used to calculate the relevant Bragg-angle for Monochromator. Default to 2.567 Angstrom
2. Source with 3 Maxwellians
3. 10 m flight path with 4 cm wide, 8 cm tall neutron guide
4. Single slab PG Monochromator, rotated according to calculated Bragg angle
5. 2 m distance to Sample (further rotated to be in the scattered beam from the mono
6. Powder sample radius 4 mm, 70 mm tall cylinder, material $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$
7. Banana-shaped detector (cylindrical cut) 10 cm tall, radius 1.5 m from sample, to record 10-180 degrees on the "high-resolution" side of the sample





“Build me a powder diffractometer ” – response:

Very detailed, comprehensive, instrument-inputs for all ‘numbers’ – don’t read this :-D

Reactor_PowderDiffractometer.instr is ready. It builds a constant-wavelength reactor powder diffractometer in the style of PSI's DMC / ILL's D2B/D20:

- **lambda** (default 2.567 Å) drives the Bragg-angle calculation in INITIALIZE: $\theta_M = \text{asin}(\text{mono_Q} \cdot \text{lambda} / (4\pi))$, using $\text{mono_Q} = 1.8734 \text{ \AA}^{-1}$ for PG(002). At the default wavelength this gives $\theta_M \approx 22.5^\circ$, $2\theta_M$ take-off $\approx 45^\circ$.
- **Source:** Source_Maxwell_3 with the canonical three-Maxwellian parameter set from McStas's own source examples ($T_1=216.8 \text{ K}$, $T_2=33.9 \text{ K}$, $T_3=16.7 \text{ K}$ with matching intensities) as a representative reactor-moderator spectrum. The simulated band is narrowed to $\text{lambda} \pm \text{dlambda}$ for statistics efficiency (a standard trick for fixed-wavelength instruments — widen dlambda to see the full spectrum).
- **Guide:** 10 m, 4 cm × 8 cm, m=2 supermirror.
- **Monochromator:** single-slab Monochromator_flat (PG mosaic, 5×5 cm), placed on an Arm rotated by the computed θ_M ; a second arm downstream picks up the full $2 \cdot \theta_M$ deflection.
- **Sample:** PowderN with Na2Ca3Al2F14.laz (the standard McStas NAC reflection data file — same one used in the real DMC instrument file), radius 4 mm, height 70 mm, wrapped in a SPLIT for better detector statistics.
- **Detector:** Monitor_nD in "banana, theta" mode, radius 1.5 m, height 10 cm, covering 10–180°, placed on the sample arm and rolled 180° about the beam axis — the same construction used in the real PSI_DMC instrument to put the bank on the focusing/high-resolution side matching the monochromator's take-off direction.

Everything is exposed as instrument parameters (guide length/size, monochromator size/mosaic, distances, sample and detector geometry) so you can retune it.

I validated this against the real thing rather than just from memory: I pulled the actual .comp source files for every component used

(Source_Maxwell_3, Guide, Arm, Monochromator_flat, PowderN, Monitor_nD, PSD_monitor, L_monitor, Progress_bar) from the McCode GitHub repo, and parsed the instrument file with mccode-antlr (the McStas grammar/compiler front-end) against those real definitions. It resolved all 14 components and every parameter name cleanly with no errors; I also confirmed the checker actually catches mistakes by deliberately introducing a typo'd parameter and an undefined component reference, both of which it flagged. Full C-code generation itself was blocked only by a sandboxed network path (github.com was reachable via raw.githubusercontent.com but not directly), which is unrelated to the instrument's own correctness. I also cross-checked the file against McStas's real PSI_DMC.instr example (same default wavelength, same detector-radius convention, same rotation idiom for the "high-resolution side"), fetched live from the repo. Sources: ...

Example 1



“Build me a powder diffractometer ” – response:

```
*****
* McStas instrument: Reactor_PowderDiffractometer
*
* %Identification
* Written by: Claude (Anthropic), for Peter
* Date: 2026-08-23
* Origin: generic reactor source
* %INSTRUMENT_SITE: Templates
*
* %Description
* Generic constant-wavelength (CW) powder neutron diffractometer for
* reactor source, in the style of instruments such as PSI's DMC or I
* D2B/D20. A triple-Maxwellian reactor source feeds a 10 m straight
* a single-slab pyrolytic graphite (PG(002)) flat monochromator sele
* wavelength and defines the Bragg take-off angle, and a cylindrical
* "banana" detector bank records the powder diffraction pattern from
* sample.
*
* The monochromator is rotated automatically from the requested wave
* 'lambda' using Bragg's law (theta_M = asin(mono_Q*lambda/(4*PI))).
* sample and detector bank are placed on the arm that carries this
* rotation, i.e. in the beam reflected by the monochromator, on the
* rotational side as the monochromator take-off ("high-resolution" /
* focusing side, following the same convention used in the McStas PS
* example instrument).
*
* The source spectrum uses the canonical three-Maxwellian parameter
* (T1=216.8 K, T2=33.9 K, T3=16.7 K with matching intensities) that
* with McStas' own Source_Maxwell_3 test/example instruments, as a
* physically reasonable stand-in for a real reactor moderator spectr
* Because this is a fixed-wavelength (CW) instrument, the simulated
* band is narrowed to +/- dlambda_pct % around lambda for statistics
* efficiency; widen 'dlambda_pct' (or edit Lmin/Lmax) to inspect the
* source spectrum.
*
* The powder sample (Na2Ca3Al2F14, "NAC") has many active reflection
* this wavelength, so PowderN's 'd_phi' vertical-focusing is used to
* restrict the simulated Debye-Scherrer cones to the detector height
* is computed automatically from det_yheight/det_radius), and 'tth_sign' is
* fixed to steer scattering onto the same side the detector bank actually
* occupies, instead of wasting half the statistics on the far side. A
* SPLIT of 58 after the sample is a good default multiplicity for this
* sample/wavelength combination.
*
* %Example: lambda=2.567 Detector: Detector_I=1e3
*
* %Parameters
* lambda: [AA]      Wavelength selected by the monochromator (sets Bragg
* mono_Q: [AA-1]    Scattering vector of the monochromator reflection (
* dlambda_pct: [%]  Half-width of the simulated source band around lamb
* L_source_guide: [m] Distance from source (moderator face) to guide entr
* L_guide: [m]      Guide length
```

```
COMPONENT mono_axis = Arm()
AT (0, 0, L_guide_mono) RELATIVE PREVIOUS

COMPONENT mono_rotation = Arm()
AT (0, 0, 0) RELATIVE mono_axis
ROTATED (0, theta_M, 0) RELATIVE mono_axis

COMPONENT mono = Monochromator_flat(
  zwidth = mono_zwidth, yheight = mono_yheight,
  mosaich = mono_mosaic, mosaicv = mono_mosaic,
  r0 = mono_r0, Q = mono_Q)
AT (0, 0, 0) RELATIVE mono_rotation

/* Output arm: carries the full 2*theta_M deflection of the reflected beam */
COMPONENT mono_out = Arm()
AT (0, 0, 0) RELATIVE mono_rotation
ROTATED (0, theta_M, 0) RELATIVE mono_rotation

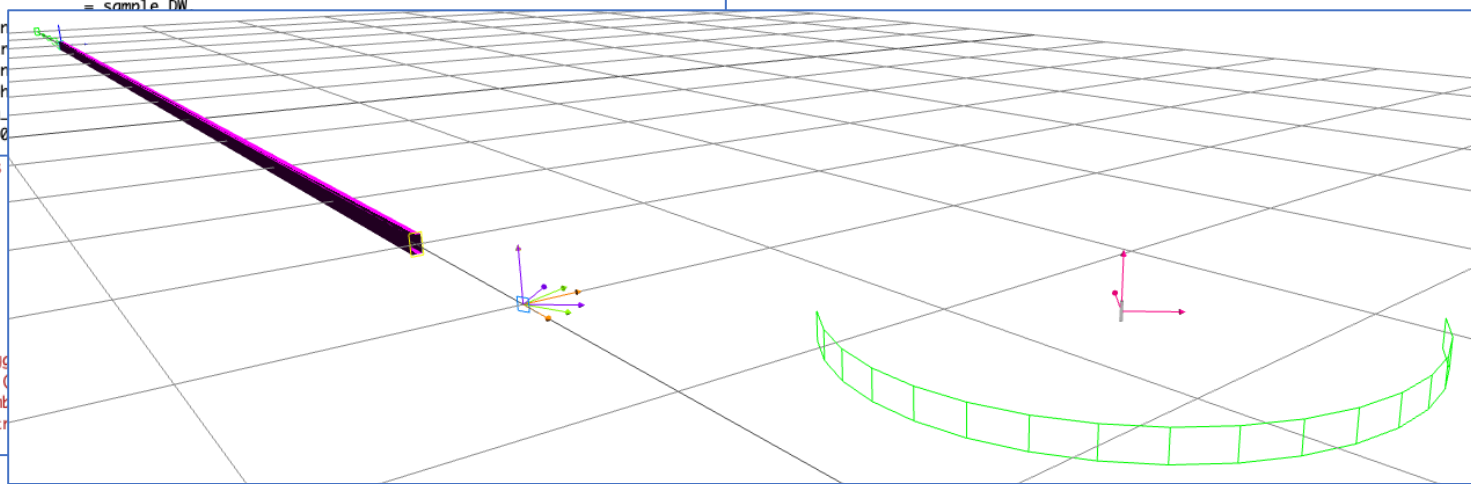
* -----
* Sample position, 2 m down the beam reflected by the monochromator
* (i.e. rotated by 2*theta_M into the scattered beam).
* -----
COMPONENT sample_arm = Arm()
AT (0, 0, L_mono_sample) RELATIVE mono_out

SPLIT sample_split COMPONENT sample = PowderNC
  radius      = sample_radius,
  yheight     = sample_yheight,
  reflections  = "Na2Ca3Al2F14.laz",
  pack        = sample_pack,
  DW          = sample_DW

p_in
p_tr
p_in
d_phi
tth
AT (0, 0, 0) RELATIVE sample
```

Simulation from prompt 1 – junk!

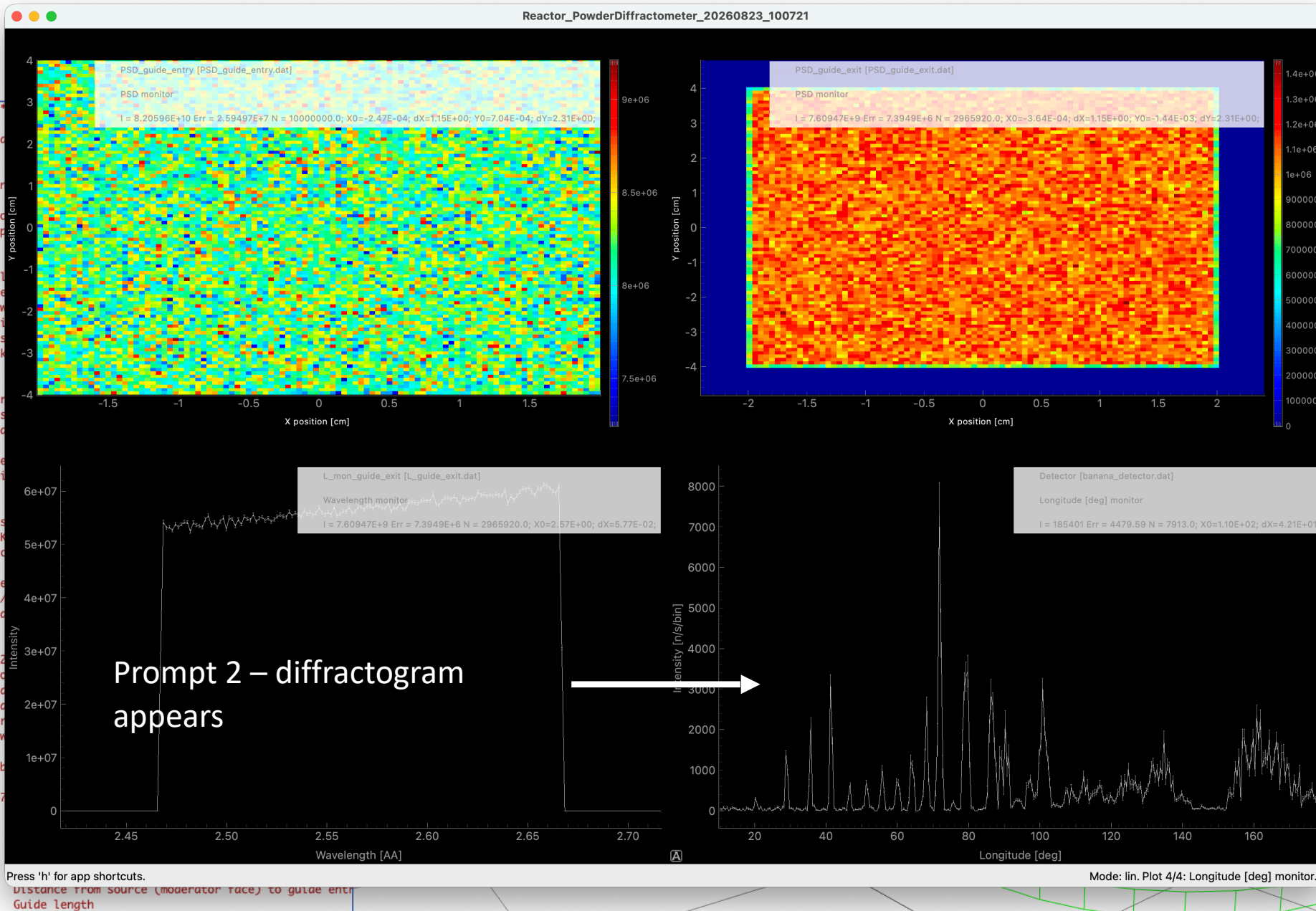
(My fault, the detector picked up direct beam from the Mono...)



Example 1



“Bu

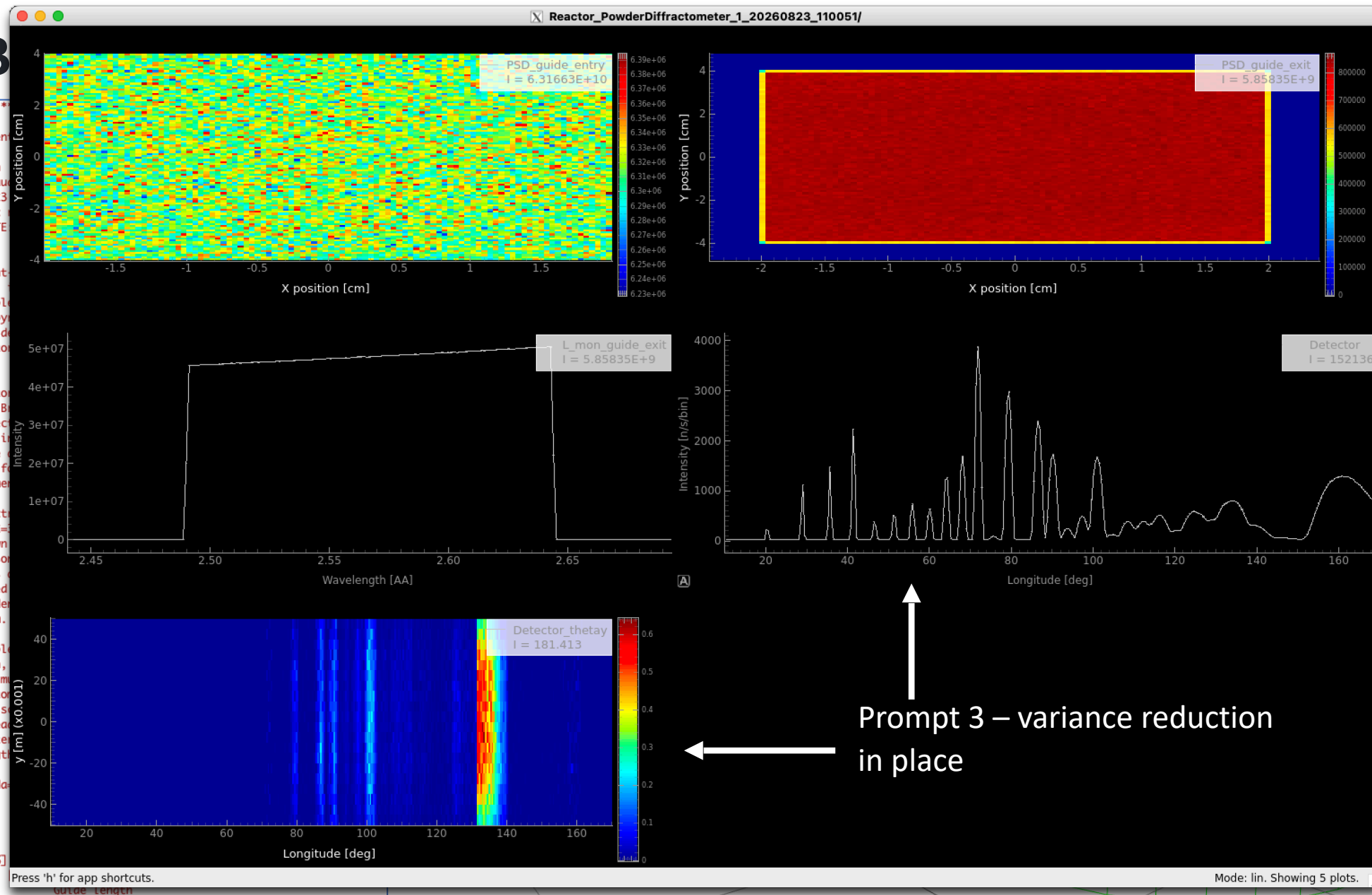


Example 1



"B"

```
*****
* McStas instrument
*
* %Identification
* Written by: Claude
* Date: 2026-08-23
* Origin: generic
* %INSTRUMENT_SITE
*
* %Description
* Generic constant
* reactor source,
* D2B/D20. A triple
* a single-slab py
* wavelength and d
* "banana" detecto
* sample.
*
* The monochromator
* 'lambda' using B
* sample and detec
* rotation, i.e. i
* rotational side
* focusing side, f
* example instrum
*
* The source spect
* (T1=216.8 K, T2=
* with McStas' own
* physically reason
* Because this is
* band is narrowed
* efficiency; wider
* source spectrum.
*
* The powder sample
* this wavelength,
* restrict the sim
* is computed auto
* fixed to steer s
* occupies, instea
* SPLIT of 58 after
* sample/wavelength
*
* %Example: lambda
*
* %Parameters
* lambda: [AA]
* mono_Q: [AA-1]
* dlambda_pct: [%]
* L_source_guide:
* L_guide: [m]
```



Prompt 3 – variance reduction
in place





Build me a model of ILL XX

Prompt 1:

Try building me an instrument that doesn't exist ready-made in McStas, ILL D17

<https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d17/>

Prompt 2:

Can you make me an ILL_D3 model? Use the information in

<https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/>

For the Source term (hot neutrons) I believe that either Source_gen or Source_Maxwell_3 has a listing of relevant parameters.

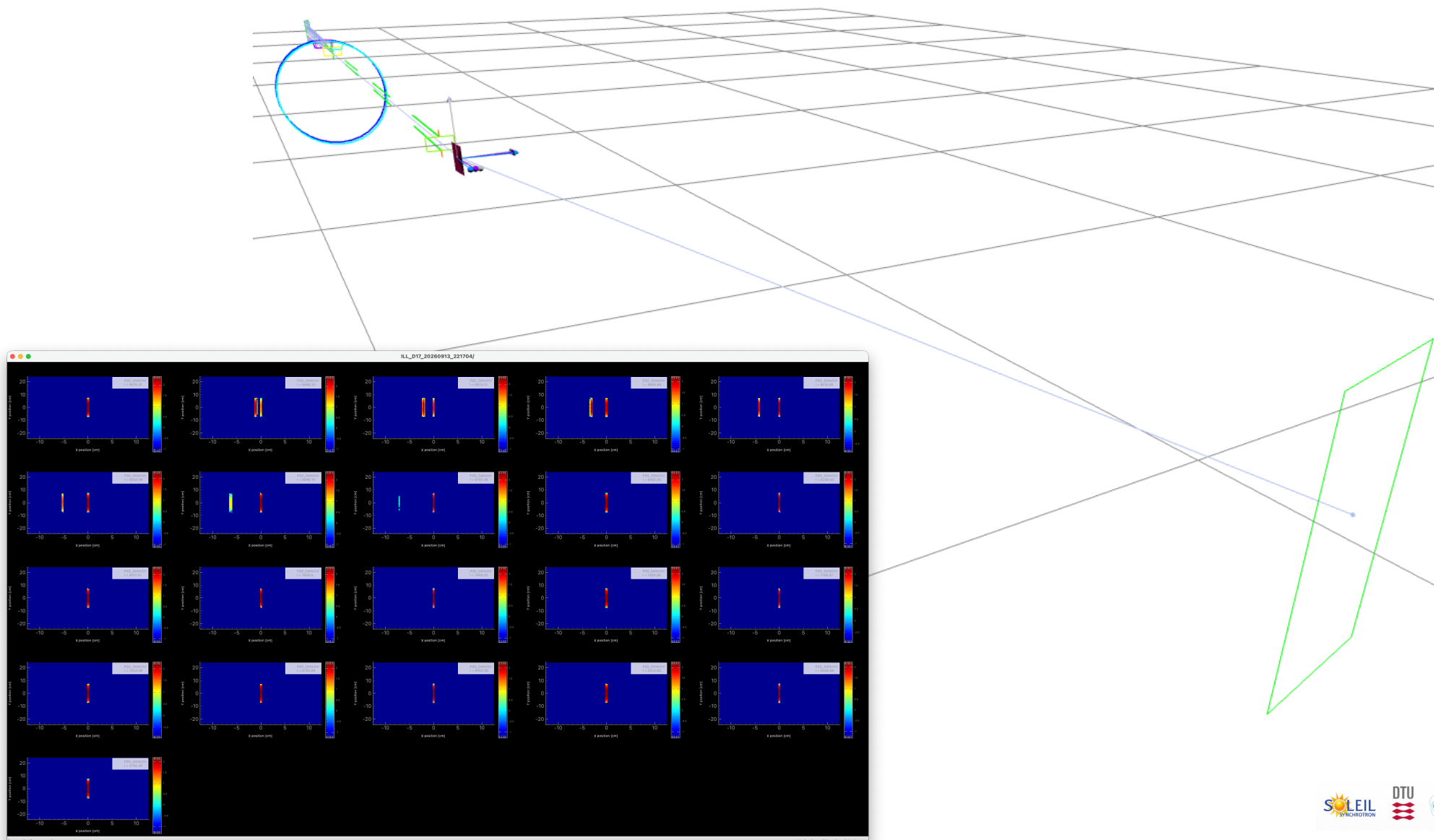
There are multiple Monochromator-options on this instrument, Monochromator_pol may be useful for the polarising mono (partly a phenomenological model), whereas the non-polarised may be simulated using either Monochromator_curved (phenomenological model) or NCrystal_sample.

If you find a design-oriented publication that would be useful for certain numbers I can probably fetch you that for round 2.



Example 2

Build me a model of ILL D17





Build me a model of ILL D3

```
*
* Sources: https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/
*           https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/characteristics/
*           https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/how-it-works/spherical-polarimetry-with-cryopad
*
* %Parameters
* mono_type: [str]    "Heussler" (polarising, high flux), "CoFe" (polarising, better res), or "Cu" (non-polarising)
* lambda: [AA]        Incident wavelength (0.42-0.843 Angs on the real instrument)
* THETA_M: [deg]       Monochromator take-off half-angle. 0 (default): derive from lambda and DM. Real range: 3.5-19 deg
*                       (i.e.  $7 < 2 * THETA\_M < 38$ )
* use_ncrystal: [1]    Cu branch only: 0 (default) Monochromator_curved mosaic model, 1 NCrystal_sample with
ncrystal_mono_cfg
* ncrystal_mono_cfg: [str] NCrystal cfg string for the Cu monochromator, used only if use_ncrystal=1
* mono_Rup: [1]        [assumed] Polarising monochromator reflectivity for spin-up
* mono_Rdown: [1]      [assumed] Polarising monochromator reflectivity for spin-down
* mono_mosaic: [arc min] Mosaic spread used for the Cu (Monochromator_curved) and polarising crystals
* L1: [m]              [assumed] Source-monochromator distance
* L2: [m]              [assumed] Monochromator-sample distance
* L3: [m]              [assumed] Sample-detector distance
* flip: [1]            0: spin-up (non-spin-flip) incident beam. 1: spin flipped (spin-flip channel) - toggles an ideal Mezei flipper
* guide_field: [T]     [assumed] Vertical guide field maintaining the polarisation quantisation axis between monochromator
and analyser
* sample_engine: [str] "vanadium" (isotropic incoherent calibration sample), "sx" (Single_crystal, supply 'reflections'/UB
matrix), "ncrystal" (NCrystal_sample, supply ncrystal_sample_cfg), or "none" (direct beam)
* sample_diam: [m]     Sample diameter (<= 0.012 m on the real instrument)
* sample_height: [m]   Sample height (<= 0.012 m on the real instrument)
* reflections: [str]   sample_engine="sx" only: reflection list file for Single_crystal
* ncrystal_sample_cfg: [str] sample_engine="ncrystal" only: NCrystal cfg string for the sample
* analyzer_type: [str] "ideal" (PolAnalyser_ideal, vertical axis), "he3" (He3_cell), or "none" (no polarisation analysis - forced
when mono_type="Cu")
* he3_p3he: [1]        [assumed] 3He cell polarisation for analyzer_type="he3" (published range ~0.70-0.85)
* det_2theta: [deg]    Detector arm scattering angle 2theta
* verbose: [1]         Print the derived instrument configuration. 0 to be quiet
```

```
* KNOWN SIMPLIFICATIONS / THINGS NOT MODELLED (please tighten with real
* engineering numbers before quantitative use):
* - CRYOPAD's zero-field spherical polarimetry (rotating nutator fields,
* Meissner shields) is not modelled at all; only a single-flipper,
* fixed-vertical-axis longitudinal polarisation analysis is implemented.
* - The magnetic (spin-dependent) part of the sample's scattering cross
* section is not modelled - none of the three 'sample_engine' choices
* computes a magnetic structure factor, so the flipping ratio you get
* out of this model reflects only the instrument's own polarising/
* analysing efficiency (Rup/Rdown, He3 cell polarisation, flipper),
* not real sample physics. That would need a bespoke sample component.
* - L1 (source-monochromator), L2 (monochromator-sample) and L3
* (sample-detector) distances, the monochromator crystal size, and the
* guide-field strengths are NOT published on the ILL website and are
* set here to plausible placeholder values, flagged [assumed] in
* DECLARE/the component calls; only DM values, the 7-38 deg take-off
* range, the 0.42-0.843 Angs band, the per-crystal flux and the 12 mm
* sample size are taken from the published characteristics page.
* - The hot H4 source is modelled as a single ~2273 K (2000 C) Maxwellian
* (Source_gen), not the real multi-term fit used in ILL's yellow-book
* source characterisation; harmonic-contamination filters (Er, Ir, Hf,
* Rh, In foils) are not modelled, so higher-order lambda/2, lambda/3
* contamination is absent here.
* - Rup/Rdown for the polarising crystals and the He3 cell polarisation
* are set to plausible round numbers (see 'mono_Rup'/'mono_Rdown'/'
* 'he3_p3he'), not measured values.
```

Heussler-mono, Cu-mono, flippers, fields.... :-O
Not perfect but absolutely relevant start!



GPT-6 Astra medium

Common conclusion for all 3 cases!



Often provides a good
starting point...

Ensure to

- * Prompt in detailed
fashion

- * Absolutely **VALIDATE** the
results!



Thank you

Questions?

mcstas.org

mcxtrace.org



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